

# MA1512 TUTORIAL 4

Week 12 (8 Apr – 12 Apr)

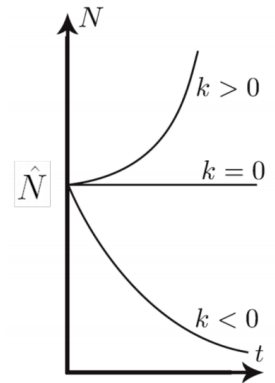
## KEY CONCEPTS – CHAPTER 3 MATHEMATICAL MODELLING

### Malthus Model of Population:

$$\frac{dN}{dt} = (B - D)N = kN$$

- Solution:  $N(t) = \hat{N}e^{kt}$ , where  $\hat{N}$  is the initial population

**Question** How do different values of  $k$  affect the prediction of population growth using the Malthus Model?



### Logistic Growth Model:

**Assumption:**  $D \sim N, D = sN$

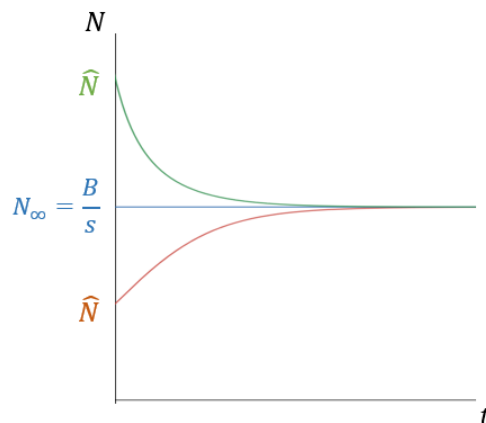
$$\frac{dN}{dt} = (B - D)N = (B - sN)N = -sN^2 + BN$$

- A logistic growth model has (1) increasing initial growth; (2) slower growth as saturation steps in; (3) growth stops upon maturation.
- Solution:

$$N = \frac{N_{\infty}}{1 + \left(\frac{N_{\infty}}{\hat{N}} - 1\right)e^{-Bt}}$$

where  $N_{\infty} = \frac{B}{s}$  (carrying capacity/logistic population) and  $N(0) = \hat{N}$ .

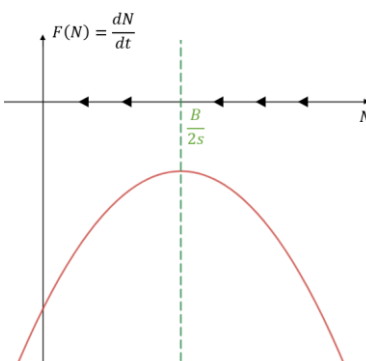
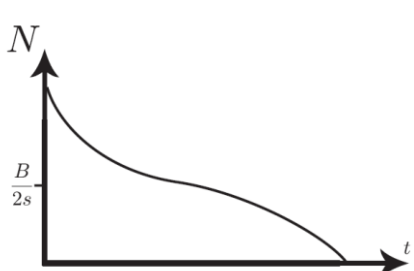
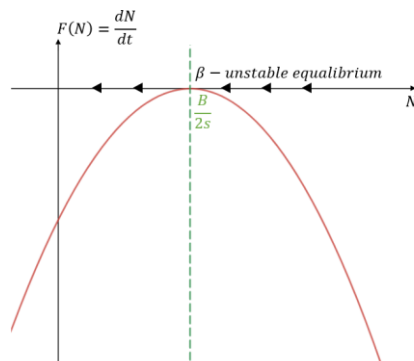
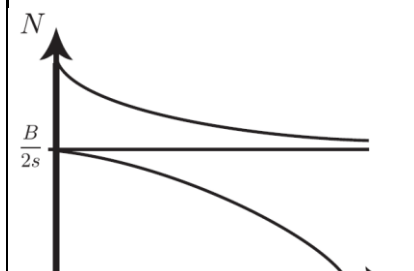
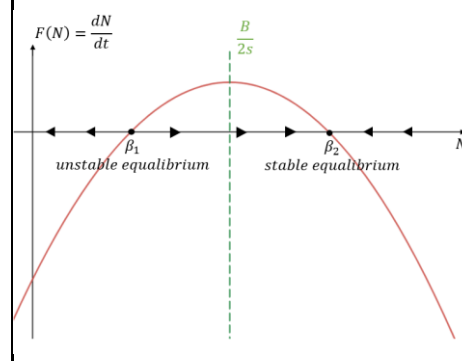
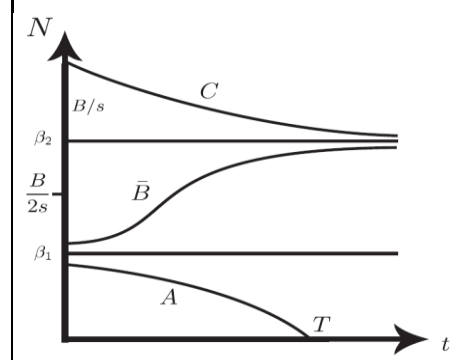
- The graph of  $N$  against  $t$  is shown below, with different initial starting values of  $N_0$ .



### Harvesting Model:

$$\frac{dN}{dt} = BN - sN^2 - E$$

To find equilibrium solutions, we consider  $BN - sN^2 - E = 0$ . The discriminant  $B^2 - 4sE$  is of concern, as it will tell us the number of equilibrium solutions we will get, as well as the behaviour of  $N$  with time  $t$ .

| Case 1: $B^2 - 4sE < 0 \Leftrightarrow E > \frac{B^2}{4s}$                                                                                                              | Case 2: $B^2 - 4sE = 0 \Leftrightarrow E = \frac{B^2}{4s}$                                                                                                                             | Case 3: $B^2 - 4sE > 0 \Leftrightarrow E < \frac{B^2}{4s}$                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br> | <br>                | <br>                                                                                |
| <p><u>Extinction!</u></p> <p><math>\frac{dN}{dt} &lt; 0, N \text{ decrease}</math></p>                                                                                  | <p><u>Extinction!</u></p> <p><math>0 &lt; x &lt; \beta: \frac{dN}{dt} &lt; 0, N \text{ decrease}</math></p> <p><math>x &gt; \beta: \frac{dN}{dt} &lt; 0, N \text{ decrease}</math></p> | <p><math>0 &lt; x &lt; \beta_1: \frac{dN}{dt} &lt; 0, N \text{ decrease}</math></p> <p><math>\beta_1 &lt; x &lt; \beta_2: \frac{dN}{dt} &gt; 0, N \text{ increase}</math></p> <p><math>x &gt; \beta_2: \frac{dN}{dt} &lt; 0, N \text{ decrease}</math></p> |

### KEY CONCEPTS – CHAPTER 4 LAPLACE TRANSFORM

- Laplace transform  $L(f)$  is a mapping  $L$  which maps a function  $f(t)$  to a function  $F(s)$ , where  $F(s)$  is given by

$$L(f(t)) = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

- The Laplace transform and inverse Laplace's transform has the **linearity** property ( $\alpha$  and  $\beta$  are constants):

$$L(\alpha f + \beta g) = \alpha L(f) + \beta L(g) \quad L^{-1}(\alpha f + \beta g) = \alpha L^{-1}(f) + \beta L^{-1}(g)$$

• Solving **Initial** Value Problems

**Step 1** Perform Laplace transform on the DE (in terms of  $t$ ) and transform it into an **algebraic** subsidiary equation.

**Step 2** Substitute in the appropriate initial values from the problem and make  $L(f)$  the subject of the formula (in terms of  $s$ ).

**Step 3** Perform inverse Laplace transform on the resultant equation to obtain the solution (in terms of  $t$ ) for the differential equation.

• The standard Laplace transforms and its inverses are presented in the table below.

|                                                                 |                                                                              |
|-----------------------------------------------------------------|------------------------------------------------------------------------------|
| $L(k) = \frac{k}{s}, k \in \mathbb{R}$                          | $L^{-1}\left(\frac{k}{s}\right) = k, k \in \mathbb{R}$                       |
| $L(t^n) = \frac{n!}{s^{n+1}}$                                   | $L^{-1}\left(\frac{1}{s^n}\right) = \frac{t^{n-1}}{(n-1)!}$                  |
| $L(e^{at}) = \frac{1}{s-a}, s > a$                              | $L^{-1}\left(\frac{1}{s-a}\right) = e^{at}$                                  |
| $L(\cos at) = \frac{s}{s^2 + a^2}, s > 0$                       | $L^{-1}\left(\frac{s}{s^2 + a^2}\right) = \cos at$                           |
| $L(\sin at) = \frac{a}{s^2 + a^2}, s > 0$                       | $L^{-1}\left(\frac{a}{s^2 + a^2}\right) = \sin at$                           |
| $L(\cosh at) = \frac{s}{s^2 - a^2}, s >  a $                    | $L^{-1}\left(\frac{s}{s^2 - a^2}\right) = \cosh at$                          |
| $L(\sinh at) = \frac{a}{s^2 - a^2}, s >  a $                    | $L^{-1}\left(\frac{a}{s^2 - a^2}\right) = \sinh at$                          |
| $L(f(t-a) \cdot u(t-a)) = e^{-as} \cdot F(s)$                   | $L^{-1}(e^{-as}F(s)) = f(t-a) \cdot u(t-a)$                                  |
| $L(\delta(t-a)) = e^{-as}$                                      | $L^{-1}(e^{-as}) = \delta(t-a)$                                              |
| $L(y') = sL(y) - y(0)$                                          |                                                                              |
| $L(y'') = s^2L(y) - sy(0) - y'(0)$                              |                                                                              |
| $L\left(\int_0^t f(\tau) d\tau\right) = \frac{1}{s}L(f), s > 0$ |                                                                              |
| Frequency-Shifting ( $s$ -shifting):                            | Time-shifting ( $t$ -shifting):                                              |
| $L(e^{ct}f(t)) = F(s-c)$                                        | $L^{-1}(e^{-as}F(s)) = f(t-a) \cdot u(t-a)$                                  |
| $L(e^{ct}t^n) = \frac{n!}{(s-c)^{n+1}}$                         | $L^{-1}\left(\frac{1}{(s-c)^n}\right) = \frac{e^{ct}t^{n-1}}{(n-1)!}$        |
| $L(e^{ct}\cos \omega t) = \frac{s-c}{(s-c)^2 + \omega^2}$       | $L^{-1}\left(\frac{s-c}{(s-c)^2 + \omega^2}\right) = e^{ct}\cos \omega t$    |
| $L(e^{ct}\sin \omega t) = \frac{\omega}{(s-c)^2 + \omega^2}$    | $L^{-1}\left(\frac{\omega}{(s-c)^2 + \omega^2}\right) = e^{ct}\sin \omega t$ |

**REMARK** Unit step function:  $u(t - a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$ . Thus,  $g(t)[u(t - a) - u(t - b)] = \begin{cases} 0, & t < a \\ g(t), & a < t < b \\ 0, & t > b \end{cases}$ .

**REMARK** The Dirac Delta function has properties  $\delta(t - a) = 0$  for  $t \neq a$  and an infinitely large magnitude (height) at the time  $t = a$ . Also, it has the property  $\int_{-\infty}^{\infty} \delta(t - a) dt = 1$  and  $\int_{-\infty}^{\infty} \delta(t - a)g(t) dt = g(a)$ .

## TUTORIAL PROBLEMS

### Question 1

The bacteria in a certain culture number 10000 initially. Two and a half hours later there are 11000 of them. Assuming a **Malthus model**, how many bacteria will there be 10 hours after the start of the experiment? How long will it take for the number to reach 20000?

#### Solutions

Since we are assuming the **Malthus model**, the following DE models the above problem:

$$\frac{dN}{dt} = kN$$

The solution (verify it) is given by (with 2 unknowns  $N_0$  and  $k$ ):

$$N(t) = \hat{N}e^{kt}$$

We are given two conditions:  $N(0) = 10000$  and  $N\left(\frac{5}{2}\right) = 11000$ . Substituting into the solution above and solving simultaneously, we get  $\hat{N} = 10000$  and  $k = \frac{2}{5} \ln 1.1 \approx 0.0381$ .

**Question** The sign of  $k$  is positive. Is it reasonable, when compared to the ODE obtained?

To find the number of bacteria after 10 hours, we let  $t = 10$ .

$$N(10) = 10000e^{(0.0381)(10)} \approx \mathbf{14641}$$

To find the time when the number of bacteria reaches 20000, let  $N(t) = 20000$ .

$$10000e^{0.0381t} = 20000 \Rightarrow t \approx \mathbf{18.18 \text{ hrs}}$$

### Question 2

You have 200 bugs in a bottle. Every day you supply them with food and count them. After two days you have 360 bugs. It is known that the **birth rate** for this kind of bug is **150%** per day. Assuming that the population is given by a **logistic model**, find the number of bugs after 3 days. Predict how many bugs you will have eventually.

#### Solutions

Assume the logistic model. The birth rate is given by  $B = 1.5$ . The DE is given by

$$\frac{dN}{dt} = (B - sN)N = BN - sN^2$$

$$\frac{dN}{dt} = 1.5N - sN^2$$

Directly use the formula for logistic population growth:

$$N = \frac{N_{\infty}}{1 + \left(\frac{N_{\infty}}{\hat{N}} - 1\right) e^{-Bt}}, \text{ where } N_{\infty} = \frac{B}{s}$$

And apply two conditions given in the question:  $\hat{N} = N(0) = 200$  and  $N(2) = 360$ ,

$$N(t) = \frac{N_{\infty}}{1 + \left(\frac{N_{\infty}}{\hat{N}} - 1\right) e^{-Bt}}$$

$$360 = \frac{N_{\infty}}{1 + \left(\frac{N_{\infty}}{200} - 1\right) e^{-1.5 \cdot 2}}$$

we can find  $N_{\infty} = B/s \approx 376$  days.

$$N(t) = \frac{N_{\infty}}{1 + \left(\frac{N_{\infty}}{\hat{N}} - 1\right) e^{-Bt}} = \frac{376}{1 + \left(\frac{376}{200} - 1\right) e^{-1.5t}}$$

To find the number of bugs when  $t = 3$ :

$$N(3) = \frac{N_{\infty}}{1 + \left(\frac{N_{\infty}}{\hat{N}} - 1\right) e^{-Bt}} = \frac{376}{1 + \left(\frac{376}{200} - 1\right) e^{-1.5 \cdot 3}} \approx 372 \text{ days}$$

To find the number of bugs eventually, we are looking at  $t \rightarrow \infty$ . This will cause the term  $Ce^{-1.5t} \rightarrow 0$  and thus,  $N(\infty) = B/s \approx 376$  days.

Remark:  $\frac{B}{s} \approx 376, B = 1.5 \rightarrow s = 0.00399$  (to be used in the next question)

### Question 3

In Question 2, let us assume that you are keeping the bugs not as a hobby, but because you are developing a new insecticide. Suppose that you remove 80 bugs per day from the bottle, and that all of these bugs die as a result of being sprayed with this insecticide. What is the limiting population in this case? **If the initial population is 100**, what is the maximum number of bugs you can put to death per day without causing the population to die out?

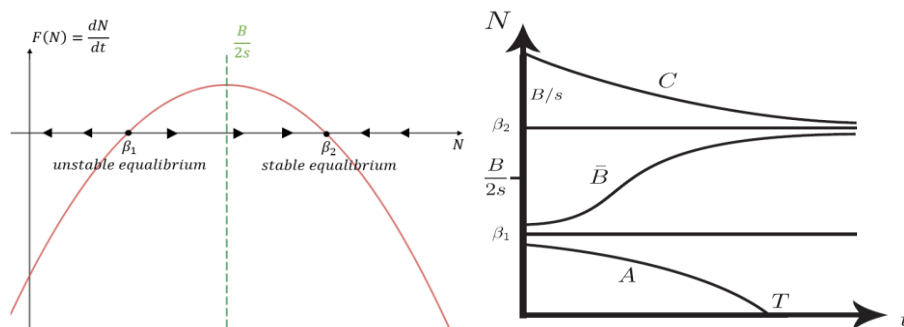
### Solutions

We will use the logistic model with harvesting.

$$\frac{dN}{dt} = -sN^2 + BN - E = -0.00399N^2 + 1.5N - 80$$

**Part 1** Find out which case of the harvesting model this question is about.

We will now assume the logistic model with harvesting,  $E = 80$ . We will calculate  $B^2 - 4sE = 0.9732 > 0$  (using  $B$  and  $s$  from Question 2) which is **Case 3**.



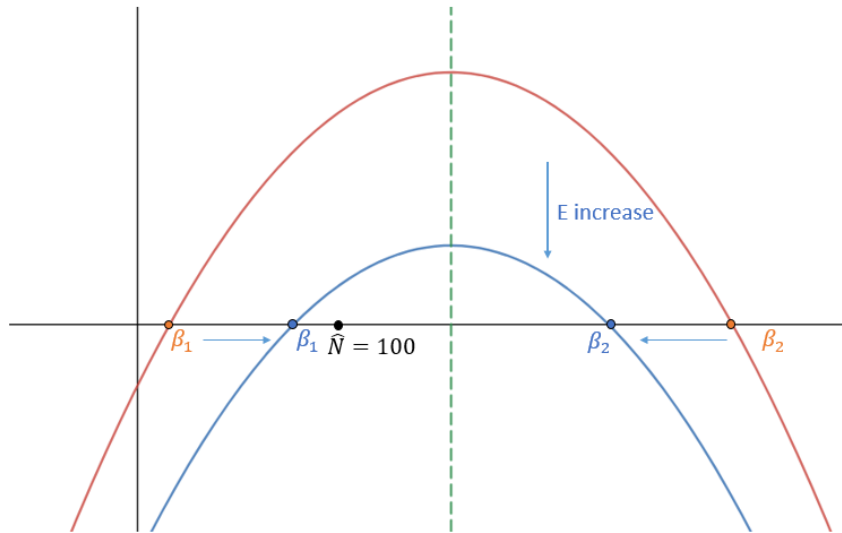
**Part 2** Classify the initial data in order to find the limiting population.

First, we need to find the values of  $\beta_1$  and  $\beta_2$ , by solving  $-0.00399N^2 + 1.5N - 80 = 0$ . Thus, we will obtain  $\beta_1 = 64.3472$  and  $\beta_2 = 311.5926$ . Since the initial population is  $N(0) = 200$  and that  $\beta_1 < N(0) < \beta_2$ , the limiting population in this case will be  $\beta_2 = 311.5926 \approx \mathbf{312}$ .

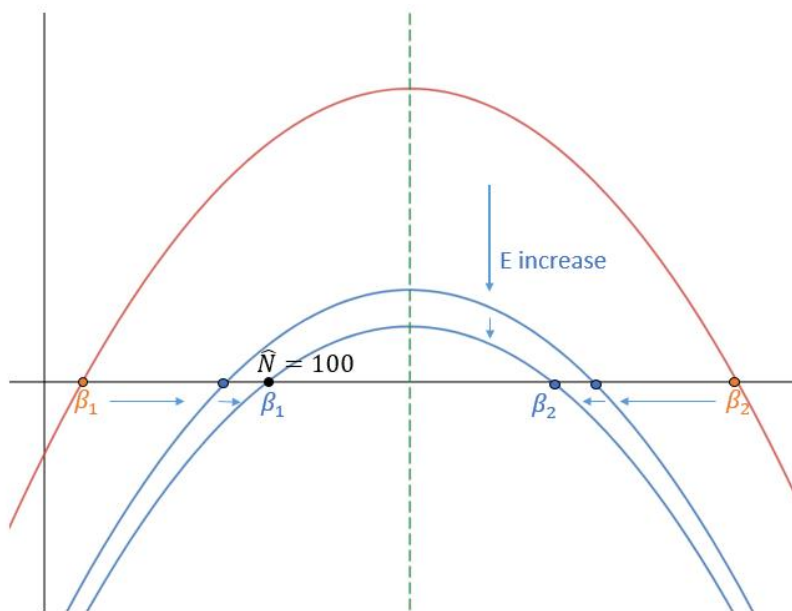
**Part 3** Find the maximum number of bugs you can kill without causing population to die out.

$$\frac{dN}{dt} = -sN^2 + BN - E$$

When  $E$  increases, the parabola shifts down:



For the population to not die out, we need  $\hat{N} \geq \beta_1$ . Thus, we should keep shifting the curve down till  $\beta_1$  coincides with  $\hat{N}$  (limiting case):



The limiting case will be when  $\hat{N} = \beta_1 = 100$ . Now, we are trying to find the value of  $E$  in the equation  $1.5N - 0.00399N^2 - E = 0$ , where  $\beta_1 = 100$  is a solution to this quadratic equation.

$$\begin{aligned} 1.5N - 0.00399N^2 - E &= 0 \\ 1.5(100) - 0.00399(100)^2 - E &= 0 \\ E &= 150 - 39.9 \approx \mathbf{110.1} \end{aligned}$$

**Question 4**

The sandhill crane is a beautiful Canadian bird with an unfortunate liking for farm crops. For many years the cranes were protected by law, and eventually they settled down to a logistic equilibrium population of 194,600 with birth rate per capita 9.866% per year. Eventually the patience of the farmers was exhausted and they managed to have the hunting ban lifted. The farmers happily shot 10000 cranes per year, which they argued was reasonable enough since it only represents about 5% of the original population. Show that the sandhill crane is doomed.

Solutions

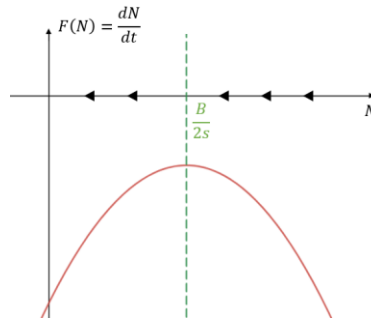
Since the logistic equilibrium  $\frac{B}{s} = 194600$  and that  $B = 0.09866$ , we get  $s = 5.07 \times 10^{-7}$ .

Consider the logistic model with harvesting  $E = 10000$ .

$$\frac{dN}{dt} = -sN^2 + BN - E$$

Calculate  $B^2 - 4sE$  to determine the case of the harvesting model.

Since  $B^2 - 4sE = -0.0105 < 0$ , this is **Case 1**, where the population of birds will decrease to zero eventually, regardless of the initial population.

Alternative method:

Maximum hunting rate is

$$\frac{B^2}{4s} = \frac{(0.09866)^2}{4 \times 5.07 \times 10^{-7}} = 4800$$

Since  $10000 > 4800$ , birds are doomed.

**Question 5**

Suppose that Peruvian fishermen take a fixed number of anchovies per year from an anchovy stock which would otherwise behave logistically, apart from occasional natural disasters. According to our lecture notes, any fishing rate  $\geq B^2/4s$  will be disastrous. Let us call this number  $E^*$ . The fishermen want to take as many anchovies as they **safely** can, meaning that they want the fish to be able to bounce back from a natural disaster that pushes their population down by 10%. Advise them. That is, tell them the maximum number of fish they can take, expressed as a percentage of  $E^*$ .

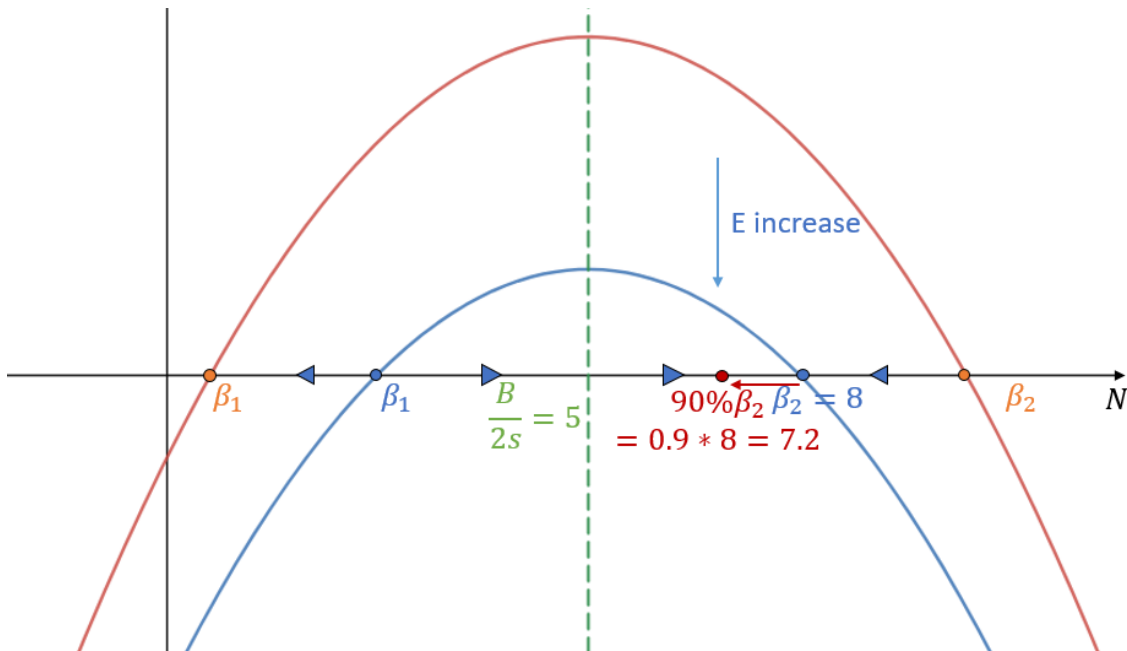
[Hint: assume that you start with the stable equilibrium population  $\beta_2$ , and compute the value of  $E$ , the harvesting rate, such that  $\beta_1$ , the **unstable** equilibrium population, becomes 90% of  $\beta_2$ .]

## Solutions

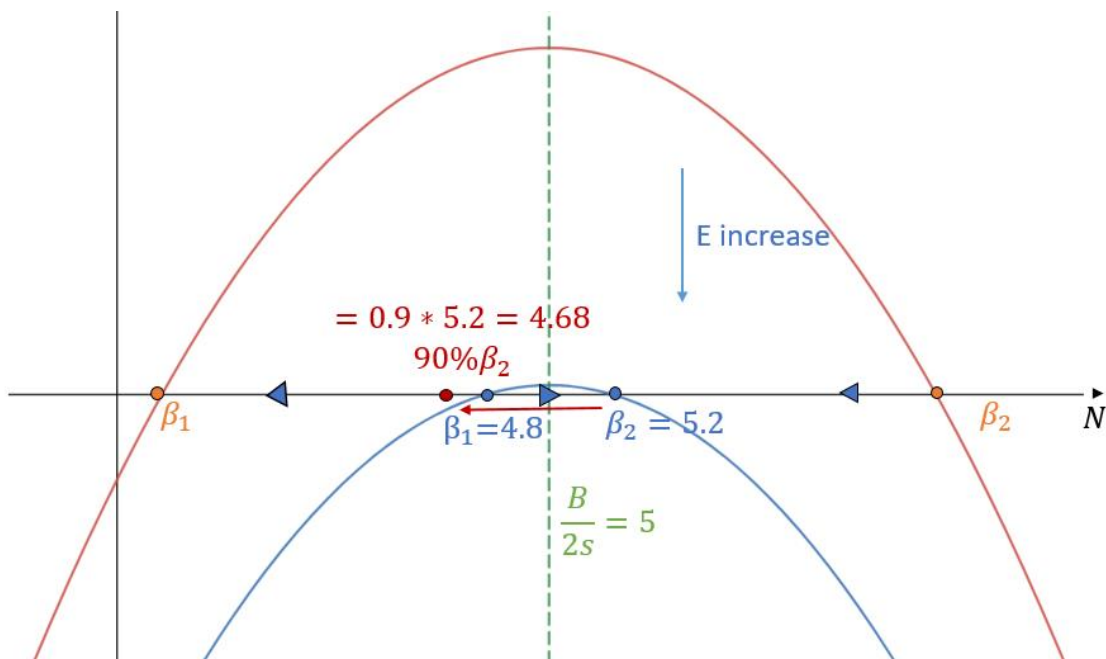
$$\frac{dN}{dt} = -sN^2 + BN - E$$

**Question:** In the following cases A & B, when a natural disaster pushes their population down by 10% (starting from equilibrium population  $\beta_2$ ), which one's population is able to bounce back to  $\beta_2$ ?

Case A



Case B



Thus, for the fish to survive a 10% downward fluctuation, in a limiting case,  $\beta_1$  coincides with  $0.9\beta_2$ .

First, we will find  $\beta_1$  and  $\beta_2$ .

$$\beta_1 = \frac{B - \sqrt{B^2 - 4sE}}{2s} \quad \beta_2 = \frac{B + \sqrt{B^2 - 4sE}}{2s}$$

Next, we need to find the value of  $E$  such that  $\beta_1$  becomes 90% of  $\beta_2$ .

$$\begin{aligned}\beta_1 &= 0.9\beta_2 \\ \frac{B - \sqrt{B^2 - 4sE}}{2s} &= 0.9 \frac{B + \sqrt{B^2 - 4sE}}{2s} \\ B - \sqrt{B^2 - 4sE} &= 0.9B + 0.9\sqrt{B^2 - 4sE} \\ 1.9\sqrt{B^2 - 4sE} &= 0.1B \\ E &= \frac{1}{4s} \left( B^2 - \frac{1}{361} B^2 \right) = \frac{360}{361} \left( \frac{B^2}{4s} \right) = \frac{360}{361} E^*\end{aligned}$$

Thus, the maximum number of fish that the fishermen can harvest is about **99.7% of  $E^*$** .

### Question 6

Find the Laplace transforms of the following functions [where  $u$  denotes the unit step function and the answers are given in brackets]:

#### Solutions

(a)  $t^2 e^{-3t}$

Using the formula  $L(t^n e^{ct}) = \frac{n!}{(s-c)^{n+1}}$ , we get  $L(t^2 e^{-3t}) = \boxed{\frac{2}{(s+3)^3}}$ .

(b)  $t \cdot u(t-2)$

Using the formula  $L(f(t-a) \cdot u(t-a)) = e^{-as} \cdot F(s)$ , we get

$$\begin{aligned}L(t \cdot u(t-2)) &= L((t-2+2) \cdot u(t-2)) \\ &= L((t-2)u(t-2)) + 2L(u(t-2)) \\ &= \frac{e^{-2s}}{s^2} + \frac{2e^{-2s}}{s} \\ &= \boxed{e^{-2s} \left( \frac{1}{s^2} + \frac{2}{s} \right)}\end{aligned}$$

What is the expression  $f(t-a)$  for each of the terms after the 2<sup>nd</sup> equality sign?

### Question 7

Find the inverse Laplace transforms of the following functions:

#### Solutions

(a)  $\frac{s}{s^2+10s+26}$

Since  $\frac{s}{s^2+10s+26} \equiv \frac{s}{(s+5)^2+1} \equiv \frac{(s+5)-5}{(s+5)^2+1}$ , we expect to use the formula  $L^{-1}\left(\frac{s-c}{(s-c)^2+\omega^2}\right) = e^{ct} \cos \omega t$  and  $L^{-1}\left(\frac{\omega}{(s-c)^2+\omega^2}\right) = e^{ct} \sin \omega t$ . Thus we get

$$\begin{aligned}L^{-1}\left(\frac{s}{s^2+10s+26}\right) &= L^{-1}\left(\frac{(s+5)-5}{(s+5)^2+1}\right) \\ &= L^{-1}\left(\frac{(s+5)}{(s+5)^2+1}\right) - 5L^{-1}\left(\frac{1}{(s+5)^2+1}\right) \\ &= e^{-5t} \cos t - 5e^{-5t} \sin t \\ &= \boxed{e^{-5t}(\cos t - 5 \sin t)}\end{aligned}$$

$$(b) \quad e^{-2s} \frac{1+2s}{s^3}$$

Since  $e^{-2s} \frac{1+2s}{s^3} = e^{-2s} \left( \frac{1}{s^3} + \frac{2}{s^2} \right)$ , define  $F(s) = \frac{1}{s^3} + \frac{2}{s^2}$ . As such, we use  $L^{-1} \left( \frac{1}{s^n} \right) = \frac{t^{n-1}}{(n-1)!}$  to get:

$$L^{-1} \left( \frac{1}{s^3} + \frac{2}{s^2} \right) = \frac{t^2}{2} + 2t = f(t)$$

Then, using  $t$ -shifting:  $L^{-1}(e^{-as}F(s)) = f(t-a) \cdot u(t-a)$  with  $a = 2$ , we have

$$\begin{aligned} L^{-1} \left( e^{-2s} \frac{1+2s}{s^3} \right) &= \left[ \frac{(t-2)^2}{2} + 2(t-2) \right] u(t-2) \\ &= \left[ \left( \frac{1}{2} t^2 - 2 \right) u(t-2) \right] \end{aligned}$$

### Question 8

Solve the following initial value problems using Laplace transforms:

#### Solutions

$$(a) \quad y' = t \cdot u(t-2) \quad y(0) = 4$$

Performing Laplace transform and assuming  $L(y) = Y(s)$ , we get

$$sY(s) - y(0) = L(t \cdot u(t-2))$$

Notice that the RHS is exactly what we are asked to evaluate in Question 6(b). Thus,

$$\begin{aligned} sY(s) - 4 &= e^{-2s} \left( \frac{1}{s^2} + \frac{2}{s} \right) \\ Y(s) &= e^{-2s} \frac{1+2s}{s^3} + \frac{4}{s} \end{aligned}$$

Now, the first term on RHS is exactly what we are asked to evaluate in Question 7(b). Thus,

$$y(t) = \left( \frac{1}{2} t^2 - 2 \right) u(t-2) + 4.$$

$$(b) \quad y'' - 2y' = 4 \quad y(0) = 1 \quad y'(0) = 0$$

Performing Laplace transform, we get

$$\begin{aligned} s^2 Y(s) - sy(0) - y'(0) - 2[sY(s) - y(0)] &= \frac{4}{s} \\ s^2 Y(s) - 2sY(s) &= \frac{4}{s} + s - 2 \\ Y(s) &= \frac{4 - 2s + s^2}{s^2(s-2)} = \frac{1}{s-2} - \frac{2}{s^2} \end{aligned}$$

Performing inverse Laplace transform (**Question** What formulas are you using now?), we get

$$y(t) = e^{2t} - 2t$$

**REMARK** This method may be simpler (and shorter) compared to the method introduced in Chapter 1 (method of undetermined coefficients or variation of parameters). Try using methods in Chapter 1 for practice.

### Question 9 [Rocket Flight]

A model rocket having initial mass  $m_0$  kg is launched vertically from the ground. The rocket expels gas at a constant rate of  $\alpha$  kg/s at a constant velocity of  $\beta$  m/s relative to the rocket. Assume that the magnitude of the gravitational force is proportional to the mass with proportionality constant  $g$ . Because the mass is not constant, Newton's second law leads to the equation

$$(m_0 - \alpha t) \frac{dv}{dt} - \alpha\beta = -g(m_0 - \alpha t),$$

where  $v = dx/dt$  is the velocity of the rocket,  $x$  is its height above the ground, and  $m_0 - \alpha t$  is the mass of the rocket  $t$  seconds after launch. If the initial velocity is zero, solve the above equation to determine the velocity of the rocket and its height above the ground for  $0 \leq t < m_0/\alpha$ .

#### Solutions

The ODE is rewritten as

$$\begin{aligned} \frac{dv}{dt} &= -g + \frac{\alpha\beta}{m_0 - \alpha t} \\ v &= -gt - \frac{\alpha\beta}{\alpha} \ln|m_0 - \alpha t| + C \end{aligned}$$

Since the initial velocity is zero, substitute initial condition  $v(0) = 0$  into equation.

$$0 = -\beta \ln|m_0| + C \Rightarrow C = \beta \ln m_0.$$

Thus,

$$\begin{aligned} v(t) &= -gt - \beta \ln(m_0 - \alpha t) + \beta \ln m_0 \\ &= -gt + \beta \ln \frac{m_0}{m_0 - \alpha t} \end{aligned}$$

Since  $v = dx/dt$  and we now solve for  $x$ :

$$\begin{aligned} \frac{dx}{dt} &= -gt + \beta \ln m_0 - \beta \ln(m_0 - \alpha t) \\ x &= -\frac{g}{2}t^2 + \beta t \ln m_0 - \beta \left[ \left( t - \frac{m_0}{\alpha} \right) \ln(m_0 - \alpha t) - t \right] + D \\ x &= -\frac{g}{2}t^2 + \beta t \ln m_0 + \frac{\beta}{\alpha} [(m_0 - \alpha t) \ln(m_0 - \alpha t) + \alpha t] + D \end{aligned}$$

$$\int \ln x \, dx = x \ln x - x$$

Since we define the starting position of model rocket to be our reference, we have initial condition  $x(0) = 0$ .

$$0 = \frac{\beta m_0}{\alpha} \ln m_0 + D \Rightarrow D = -\frac{\beta m_0}{\alpha} \ln m_0$$

Then,

$$\begin{aligned} x &= -\frac{g}{2}t^2 + \beta t \ln m_0 + \frac{\beta}{\alpha} [(m_0 - \alpha t) \ln(m_0 - \alpha t) + \alpha t] - \frac{\beta m_0}{\alpha} \ln m_0 \\ &= -\frac{g}{2}t^2 + \beta t \ln m_0 + \frac{\beta m_0}{\alpha} \ln(m_0 - \alpha t) - \beta t \ln(m_0 - \alpha t) + \beta t - \frac{\beta m_0}{\alpha} \ln m_0 \\ &= -\frac{g}{2}t^2 + \beta t + \beta t \ln \frac{m_0}{m_0 - \alpha t} + \frac{\beta m_0}{\alpha} \ln \frac{m_0 - \alpha t}{m_0} \\ &= \beta t - \frac{g}{2}t^2 + \beta t \ln \frac{m_0}{m_0 - \alpha t} - \frac{\beta m_0}{\alpha} \ln \frac{m_0}{m_0 - \alpha t} \\ \boxed{x &= \beta t - \frac{g}{2}t^2 - \frac{\beta}{\alpha} (m_0 - \alpha t) \ln \frac{m_0}{m_0 - \alpha t}} \end{aligned}$$

**Question 10**

In the harvesting model we considered in the lectures, the population will rebound if all harvesting is stopped. Unhappily, this is not always true: for some animals, if you drive their population down too low, they will have trouble finding mates, or they will be forced to breed with relatively close kin, which reduces genetic variability and hence their ability to resist disease. For such animals [for example, certain rare species of tigers] extinction will result if the population falls too low, even if all harvesting is forbidden. Biologists call this **depensation**. Show that this situation can be modelled by the ODE

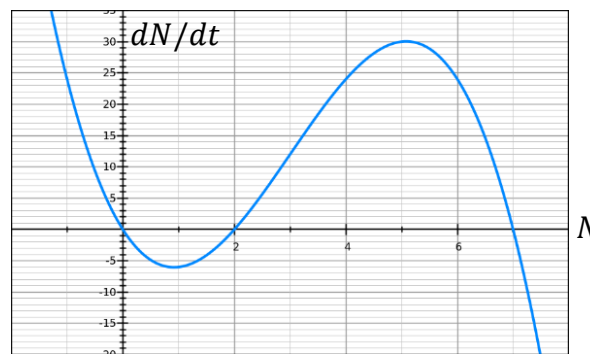
$$\frac{dN}{dt} = -aN^3 + bN^2 - cN$$

where  $N$  is the population and  $a, b$ , and  $c$  are positive constants such that  $b^2 > 4ac$ . Find the population below which extinction will occur.

Solutions

**Part I** Show that the situation can be modelled by  $\frac{dN}{dt} = -aN^3 + bN^2 - cN$ .

Depensation occurs if the population is low enough, the population will continue to fall till extinction. Translating into mathematics, it means that if  $N$  is small enough,  $\frac{dN}{dt}$  will be negative. Clearly, for  $a, b$ , and  $c$  being positive constants, the DE  $\frac{dN}{dt} = -aN^3 + bN^2 - cN$  gives negative  $\frac{dN}{dt}$  for small values of  $N$ , as seen in the graph below.



As such, this is a suitable model for the phenomenon of depensation.

**Part II** Find out the population below which extinction will occur.

For extinction to occur, the rate of change of population  $\frac{dN}{dt}$  should be decreasing till the population reaches zero. Observing the graph above, that population value we are looking for should be the smallest  $N$ -intercept other than  $N = 0$ . Hence, we need to solve

$$\begin{aligned} -aN^3 + bN^2 - cN &= 0 \\ -N(aN^2 - bN + c) &= 0 \\ N = 0 \text{ OR } N &= \frac{b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$$

Thus, the required maximum population value is  $\frac{b - \sqrt{b^2 - 4ac}}{2a}$ .

**Question** Classify the stability of each of the equilibrium solutions.

**REMARK** Notice that harvesting is not present in this whole situation. However, the behaviour of depensation is very similar to the logistic model with harvesting. Hence, this is a good model for depensation.